# Graph Neural Network Based Near-Field Detection for XL-MIMO via Dynamic Visibility Region Discovery

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***Abstract*—Abstract—Extremely large-scale multiple-input multiple-output (XL-MIMO) is a pivotal technology for 6G communications; however, it introduces severe near-field spatial non-stationarity. Consequently, individual users are effectively covered by only a localized subset of antennas, known as the visibility region (VR). Under these conditions, traditional full-array processing becomes computationally prohibitive and susceptible to noise accumulation, whereas fully-connected graph neural network (GNN) detectors suffer from severe scalability bottlenecks. To address these challenges, this paper proposes Graph-VR-Det, a graph-based near-field detection scheme. The proposed method utilizes pilot-energy distributions for dynamic VR discovery to construct sparse user-antenna bipartite graphs. By restricting the evidence processing of a sparse message passing neural network (Sparse-GNN) exclusively to valid edges, computational complexity is significantly reduced. Furthermore, a frequency-aware extension is introduced to compensate for wideband beam splitting effects at terahertz (THz) frequencies. Simulations demonstrate that the proposed Sparse-GNN achieves up to a 3.3-fold inference speedup over fully-connected networks. At an optimal sparsity level, the proposed method exhibits a graph regularization effect that filters out noise, surpassing the bit error rate performance of dense networks. The architecture also demonstrates zero-shot scalability to 1024-antenna arrays without retraining. However, in wideband scenarios, the proposed method exhibits an error floor at high signal-to-noise ratios compared to ideal genie-aided linear detectors, highlighting a trade-off between computational efficiency and high-SNR precision.**

## I. Introduction

The evolution of wireless communication networks toward the sixth generation (6G) is driven by the demand for ultra-high data rates and massive connectivity. To meet these requirements, extremely large-scale multiple-input multiple-output (XL-MIMO) operating in the sub-terahertz (sub-THz) and THz bands has emerged as a pivotal technology [1], [2]. By deploying hundreds or thousands of antenna elements, XL-MIMO provides substantial spatial degrees of freedom and array gains. However, the significant increase in the antenna array aperture, coupled with high operating frequencies, alters the electromagnetic propagation characteristics. Specifically, the conventional far-field planar wave assumption becomes invalid, as users are frequently located within the Fresnel zone of the base station (BS). Consequently, the communication paradigm shifts from far-field planar waves to near-field spherical wave propagation, which necessitates a re-evaluation of transceiver design [3], [4].

A consequence of near-field spherical wave propagation in XL-MIMO systems is the emergence of spatial non-stationarity [5]. Unlike conventional massive MIMO systems where the signal energy is uniformly distributed across the entire array, the signal from a specific user in an XL-MIMO system is effectively captured by only a localized subset of antennas. This localized subset is formally defined as the visibility region (VR) [6], [7]. The presence of VRs implies that processing the received signal across the entire antenna array is computationally inefficient and potentially detrimental to system performance. Traditional linear detection algorithms, such as zero-forcing (ZF) and minimum mean square error (MMSE) detectors, require large-scale matrix inversions. When applied to the full XL-MIMO array, these methods incur substantial computational complexity. Furthermore, incorporating antennas outside the user's VR into the detection process introduces noise accumulation, which degrades the overall detection accuracy [3].

To overcome the computational bottlenecks of traditional linear detectors, deep learning (DL) has been investigated for MIMO detection [8]. Among various DL architectures, graph neural networks (GNNs) have demonstrated effectiveness by modeling the MIMO detection problem as a message-passing process on a bipartite graph [9], [10]. Existing GNN-based detectors achieve near-optimal performance with reduced complexity compared to iterative algorithms [11], [12]. However, a technical challenge remains: state-of-the-art GNN detectors predominantly rely on fully-connected graph topologies, where every user node is connected to every antenna node. In XL-MIMO scenarios, this fully-connected assumption exhibits limited scalability, yielding a computational complexity of $\mathcal{O}(NK)$ per layer, where $N$ and $K$ are the numbers of antennas and users, respectively. As $N$ grows to the thousands, fully-connected GNNs encounter dimensionality challenges and memory overflow (out-of-memory, OOM) issues. Moreover, in wideband THz XL-MIMO systems, the near-field beam splitting effect causes the VR to shift across different subcarriers [13], [14], rendering static, fully-connected graph structures suboptimal and less capable of capturing the frequency-dependent spatial non-stationarity.

To address these challenges, this paper proposes Graph-VR-Det, a graph-based near-field detection scheme for XL-MIMO systems via dynamic VR discovery. By exploiting spatial non-stationarity, the XL-MIMO detection is modeled as a sparse message-passing problem, reducing computational overhead while maintaining detection precision. The contributions of this paper are summarized as follows:

- A dynamic, pilot-based VR discovery algorithm is pro-

posed to extract sparse user-antenna bipartite graph topologies without requiring prior channel state information (CSI). By evaluating the pilot-energy distributions, the algorithm identifies the valid VR edges for each user.
- A Sparse-GNN detector is developed, which restricts evidence processing and message passing exclusively to the discovered valid edges. This sparse activation mechanism reduces the computational complexity from scaling linearly with the total number of antennas to scaling linearly with the VR size.
- A frequency-aware feature mechanism is introduced to mitigate the wideband near-field beam splitting effect at THz frequencies. By integrating normalized frequency offsets into the edge features and constructing a shared graph topology across subcarriers, the proposed GNN compensates for frequency-dependent spatial shifts.
- Numerical evaluations demonstrate that the proposed Sparse-GNN achieves up to a 3.3-fold inference speedup over fully-connected networks. At an optimal sparsity level, the sparse graph topology induces a regularization effect that filters out noise, allowing the Sparse-GNN to surpass the bit error rate (BER) performance of dense networks. The architecture also exhibits zero-shot generalization capabilities, scaling to 1024-antenna arrays without retraining. However, it is noted that in wideband scenarios, the proposed method exhibits an error floor at high signal-to-noise ratios compared to genie-aided linear detectors, highlighting a trade-off between computational efficiency and high-SNR precision.

The remainder of this paper is organized as follows. Section II introduces the system model, detailing the near-field wideband channel and the problem formulation. Section III presents the proposed Graph-VR-Det methodology, including the dynamic VR discovery and the frequency-aware Sparse-GNN architecture. Section IV provides numerical results and complexity analyses. Finally, Section V concludes the paper.

## II. System Model

### A. Narrowband Near-Field XL-MIMO System

Consider an uplink extremely large-scale multiple-input multiple-output (XL-MIMO) communication system, where a base station (BS) equipped with an $N$-element uniform linear array (ULA) serves $K$ single-antenna users. The antennas at the BS are uniformly spaced with an inter-element distance $d = \lambda_c/2$, where $\lambda_c$ denotes the carrier wavelength. The physical coordinates of the $n$-th BS antenna are given by $\mathbf{q}_n = [0, (n-1-N/2)d, 0]^T$ for $n = 1, 2, \ldots, N$. The $K$ users are randomly distributed within the Fresnel near-field region of the BS array. The coordinates of the $k$-th user are denoted by $\mathbf{p}_k = [D_k \cos\theta_k, D_k \sin\theta_k, 0]^T$, where $D_k$ represents the radial distance from the center of the ULA to the $k$-th user, and $\theta_k \in [-\pi/2, \pi/2]$ is the corresponding azimuth angle. In the near-field regime, the traditional far-field planar wave assumption becomes invalid. Consequently, electromagnetic propagation must be characterized by an exact spherical wave model. The Euclidean distance between the $n$-th BS antenna and the $k$-th user is given by $r_{n,k} = \|\mathbf{p}_k - \mathbf{q}_n\|_2$. Accordingly, the narrowband near-field channel coefficient between the $k$-th user and the $n$-th antenna is modeled as

$$\tilde{H}_{n,k} = \frac{\lambda_c}{4\pi r_{n,k}} e^{-j\frac{2\pi}{\lambda_c} r_{n,k}}, \tag{1}$$

where the amplitude term $\frac{\lambda_c}{4\pi r_{n,k}}$ captures the free-space path loss, and the exponential term $e^{-j\frac{2\pi}{\lambda_c} r_{n,k}}$ represents the phase shift induced by the spherical wave propagation delay.

### B. Spatial Non-Stationarity and Visibility Region

Given the physically large aperture of the XL-MIMO array and the relative proximity of the users, the signal energy radiated by a specific user is not uniformly distributed across all BS antennas. Instead, the signal is predominantly captured by a localized, contiguous subset of antennas, a phenomenon referred to as spatial non-stationarity. To mathematically model this effect, a visibility region (VR) mask matrix $\mathbf{M} \in \mathbb{R}^{N\times K}$ is defined. For the $k$-th user, the signal is assumed to be visible only to a subset of antennas centered around an index $c_k \in \{1, 2, \ldots, N\}$ with a visibility radius $R_v$. The elements of the VR mask matrix are thus given by

$$M_{n,k} = \begin{cases} 1, & \text{if } |n - c_k| \le R_v, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

Incorporating spatial non-stationarity, the effective narrowband channel matrix $\mathbf{H} \in \mathbb{C}^{N\times K}$ is expressed as the Hadamard product of the VR mask matrix and the spherical wave channel matrix, i.e., $\mathbf{H} = \mathbf{M} \odot \tilde{\mathbf{H}}$, where $[\tilde{\mathbf{H}}]_{n,k} = \tilde{H}_{n,k}$. The application of the VR mask naturally induces a sparse structure in the channel matrix $\mathbf{H}$, which can be exploited to mitigate the computational complexity of subsequent signal processing algorithms.

### C. Wideband OFDM System and Near-Field Beam Splitting

To support ultra-high data rates, the system model is extended to a wideband orthogonal frequency-division multiplexing (OFDM) architecture operating in the terahertz (THz) band. Let $f_c$ denote the center frequency and $B$ denote the total system bandwidth, which is uniformly divided into $F$ orthogonal subcarriers. The frequency of the $m$-th subcarrier is given by $f_m = f_c + \frac{B}{F}(m-1-F/2)$ for $m = 1, 2, \ldots, F$. In wideband THz XL-MIMO systems, propagation characteristics become highly frequency-dependent. The wideband near-field channel coefficient for the $m$-th subcarrier is formulated as

$$H_{n,k,m} = M_{n,k} \frac{c}{4\pi f_m r_{n,k}} e^{-j\frac{2\pi f_m}{c} r_{n,k}}, \tag{3}$$

where $c$ is the speed of light. The frequency-dependent phase shift $e^{-j\frac{2\pi f_m}{c} r_{n,k}}$ induces the near-field beam splitting effect. Specifically, the spatial focusing point of the array shifts across different subcarriers, implying that the effective VR for a given user may vary with frequency. This frequency-dependent spatial non-stationarity poses a significant challenge for wideband channel estimation and signal detection, as static processing architectures fail to capture subcarrier-varying spatial features.

### D. Signal Transmission and Problem Formulation

Let $\mathbf{x}_m = [x_{1,m}, x_{2,m}, \ldots, x_{K,m}]^T \in \mathbb{C}^{K\times 1}$ denote the vector of transmitted symbols from the $K$ users on the $m$-th subcarrier. The symbols are drawn from a modulation constellation (e.g., quadrature phase-shift keying, QPSK) such that $\mathbb{E}[\mathbf{x}_m\mathbf{x}_m^H] = \mathbf{I}_K$. The received signal vector $\mathbf{y}_m \in \mathbb{C}^{N\times 1}$ at the BS on the $m$-th subcarrier is modeled as

$$\mathbf{y}_m = \mathbf{H}_m\mathbf{x}_m + \mathbf{n}_m, \tag{4}$$

where $\mathbf{H}_m \in \mathbb{C}^{N\times K}$ is the channel matrix for the $m$-th subcarrier with $[\mathbf{H}_m]_{n,k} = H_{n,k,m}$, and $\mathbf{n}_m \sim \mathcal{CN}(\mathbf{0}, \sigma^2\mathbf{I}_N)$ is the additive white Gaussian noise (AWGN) vector with noise variance $\sigma^2$.

The primary objective of the detection process is to accurately and efficiently recover the transmitted symbols $\mathbf{x}_m$ from the received signal $\mathbf{y}_m$ across all subcarriers $m = 1, \ldots, F$. Traditional linear detection algorithms, such as full-array zero-forcing (ZF) and minimum mean square error (MMSE) detectors, compute the estimated symbols as $\hat{\mathbf{x}}_m^{\text{ZF}} = (\mathbf{H}_m^H\mathbf{H}_m)^{-1}\mathbf{H}_m^H\mathbf{y}_m$ and $\hat{\mathbf{x}}_m^{\text{MMSE}} = (\mathbf{H}_m^H\mathbf{H}_m + \sigma^2\mathbf{I}_K)^{-1}\mathbf{H}_m^H\mathbf{y}_m$, respectively. However, these methods necessitate large-scale matrix inversions of size $K \times K$ and involve dense matrix multiplications that scale linearly with the total number of antennas $N$, leading to prohibitive computational complexity in XL-MIMO systems. Furthermore, processing signals from antennas outside a user's VR introduces severe noise accumulation, which degrades detection performance. To establish theoretical performance upper bounds, genie-aided VR-ZF and VR-MMSE detectors are introduced, which assume perfect prior knowledge of the true VR mask $\mathbf{M}$. These genie-aided detectors perform linear detection exclusively on the non-zero elements of the sparse channel matrix $\mathbf{H}_m$, thereby eliminating noise contributions from non-visible antennas. The objective of the proposed methodology is to design a low-complexity, scalable detection algorithm that dynamically discovers the VR and approaches the performance of the genie-aided bounds while effectively compensating for the wideband beam splitting effect.

## III. Proposed Graph-VR-Det Methodology

### A. Overall Architecture

The Graph-VR-Det framework reformulates the near-field XL-MIMO detection task as a regression problem on a dynamically constructed sparse bipartite graph. Let $\mathcal{G} = (\mathcal{V}, \mathcal{U}, \mathcal{E})$ denote an undirected bipartite graph, where $\mathcal{V} = \{v_1, v_2, \ldots, v_N\}$ represents the set of $N$ base station antenna nodes, and $\mathcal{U} = \{u_1, u_2, \ldots, u_K\}$ represents the set of $K$ user nodes. The edge set $\mathcal{E}$ defines the connectivity between antennas and users. In conventional fully-connected GNN-based detectors, $\mathcal{E}$ contains $NK$ edges, implying that every user's signal is processed by the entire array. However, due to the spatial non-stationarity inherent in near-field XL-MIMO channels, the signal energy of each user is concentrated within a localized visibility region (VR). Consequently, processing the full graph introduces severe noise accumulation from non-visible antennas and incurs prohibitive computational overhead. To address this limitation, Graph-VR-Det introduces a dynamic VR discovery mechanism to construct a sparse edge set $\mathcal{E}_{\text{sparse}} \subset \mathcal{E}$, where $|\mathcal{E}_{\text{sparse}}| = SK \ll NK$, with $S$ denoting the number of retained antennas per user. A sparse message passing neural network (Sparse-GNN) is then executed exclusively over $\mathcal{E}_{\text{sparse}}$ to iteratively refine the estimates of the transmitted symbols.

### B. Dynamic Visibility Region Discovery

The construction of the sparse graph topology relies on a pilot-based dynamic VR discovery algorithm, operating without requiring perfect prior knowledge of the channel state information (CSI) or the true VR mask $\mathbf{M}$. During the uplink training phase, orthogonal pilot sequences are transmitted by the $K$ users. Let $\mathbf{X}_p = \mathbf{I}_K$ denote the pilot matrix. The received pilot signal at the BS is given by $\mathbf{Y}_p = \mathbf{H} + \mathbf{N}_p$, where $\mathbf{N}_p$ is the additive noise matrix during the pilot phase. To identify the effective VR for each user, the receiver computes the energy of the received pilot signals. The energy matrix $\mathbf{E} \in \mathbb{R}^{N\times K}$ is calculated element-wise as $E_{n,k} = |[\mathbf{Y}_p]_{n,k}|^2$. For the $k$-th user, the algorithm selects the indices of the antennas corresponding to the $S$ largest energy values in the $k$-th column of $\mathbf{E}$. This top-$S$ selection dynamically generates a sparse binary adjacency matrix $\mathbf{A}_{\text{sparse}} \in \{0,1\}^{N\times K}$, where $[\mathbf{A}_{\text{sparse}}]_{n,k} = 1$ if the $n$-th antenna is within the discovered VR of the $k$-th user, and 0 otherwise. The non-zero entries of $\mathbf{A}_{\text{sparse}}$ define the sparse edge set $\mathcal{E}_{\text{sparse}}$. By relying on the received signal energy rather than exact channel coefficients, this approach is robust to estimation errors and effectively filters out antennas dominated by noise.

### C. Sparse Message Passing Neural Network

Given the sparse graph topology $\mathcal{E}_{\text{sparse}}$, the Sparse-GNN performs iterative evidence processing to recover the transmitted symbols. The process begins with the initialization of node and edge features. The initial feature vector for the $n$-th antenna node is defined by the real and imaginary parts of the received signal, i.e., $\mathbf{v}_n^{(0)} = [\Re(y_n), \Im(y_n)]^T \in \mathbb{R}^2$. The initial feature vector for the $k$-th user node is set to zero, $\mathbf{u}_k^{(0)} = [0, 0]^T \in \mathbb{R}^2$. The edge feature between the $n$-th antenna and the $k$-th user is initialized using the estimated channel coefficient, $\mathbf{e}_{n,k} = [\Re(H_{n,k}), \Im(H_{n,k})]^T \in \mathbb{R}^2$. These initial features are mapped to a higher-dimensional hidden space $\mathbb{R}^{d_h}$ via linear embedding layers, where $d_h$ is the hidden dimension.

The core of the Sparse-GNN consists of $L$ message passing layers. In the $l$-th layer, where $l \in \{1, 2, \ldots, L\}$, the network performs Antenna-to-User (A2U) and User-to-Antenna (U2A) updates strictly along the edges in $\mathcal{E}_{\text{sparse}}$. During the A2U phase, the message from antenna $n$ to user $k$ is computed using a multi-layer perceptron (MLP), denoted as $\text{MLP}_{\text{A2U}}$, which takes the concatenated features of the connected nodes and the edge as input:

$$\mathbf{m}_{n\to k}^{(l)} = \text{MLP}_{\text{A2U}}\left(\left[\mathbf{v}_n^{(l-1)}, \mathbf{u}_k^{(l-1)}, \mathbf{e}_{n,k}\right]\right). \tag{5}$$

The messages are then aggregated at the $k$-th user node using a scatter-add operation over the valid edges, and the user feature is updated via $\text{MLP}_\text{U}$:

$$\mathbf{u}_k^{(l)} = \text{MLP}_\text{U}\left(\left[\mathbf{u}_k^{(l-1)}, \frac{1}{S}\sum_{n\in\mathcal{N}(k)} \mathbf{m}_{n\to k}^{(l)}\right]\right), \quad (6)$$

where $\mathcal{N}(k)$ denotes the set of antennas connected to user $k$ in $\mathcal{E}_\text{sparse}$. Subsequently, during the U2A phase, messages from users to antennas are computed and aggregated similarly:

$$\mathbf{m}_{k\to n}^{(l)} = \text{MLP}_\text{U2A}\left(\left[\mathbf{u}_k^{(l)}, \mathbf{v}_n^{(l-1)}, \mathbf{e}_{n,k}\right]\right), \quad (7)$$

$$\mathbf{v}_n^{(l)} = \text{MLP}_\text{A}\left(\left[\mathbf{v}_n^{(l-1)}, \frac{1}{|\mathcal{N}(n)|}\sum_{k\in\mathcal{N}(n)} \mathbf{m}_{k\to n}^{(l)}\right]\right), \quad (8)$$

where $\mathcal{N}(n)$ is the set of users connected to antenna $n$, and $|\mathcal{N}(n)|$ is its dynamic degree. After $L$ iterations, a linear readout layer maps the final user features $\mathbf{u}_k^{(L)}$ to a two-dimensional space to predict the real and imaginary parts of the transmitted symbols $\hat{\mathbf{x}}$.

### D. Frequency-Aware Wideband Extension

To extend the architecture to wideband THz XL-MIMO systems, the near-field beam splitting effect must be addressed. Beam splitting causes the spatial focusing point, and consequently the VR, to shift across different subcarriers. Processing each subcarrier independently with a separate graph topology is computationally inefficient and ignores the underlying physical correlation of the wideband channel. Therefore, Graph-VR-Det introduces a joint frequency-domain graph mapping strategy. During the pilot phase, the energy matrix is computed for each subcarrier $m$, denoted as $\mathbf{E}_m$. The algorithm then calculates the average energy across all $F$ subcarriers, $\bar{\mathbf{E}} = \frac{1}{F}\sum_{m=1}^{F}\mathbf{E}_m$. The top-$S$ selection is performed on $\bar{\mathbf{E}}$ to generate a shared, stable graph topology $\mathcal{E}_\text{sparse}$ that is applied universally across all subcarriers.

To enable the GNN to learn and compensate for the frequency-dependent spatial shifts within this shared topology, frequency-aware edge features are introduced. For the $m$-th subcarrier, the normalized frequency offset $\Delta f_m = (f_m - f_c)/f_c$ is concatenated with the channel coefficients to form an augmented three-dimensional edge feature vector:

$$\mathbf{e}_{n,k,m} = \left[\Re(H_{n,k,m}), \Im(H_{n,k,m}), \frac{f_m - f_c}{f_c}\right]^T \in \mathbb{R}^3. \quad (9)$$

By embedding the explicit frequency information into the message passing kernel, the Sparse-GNN dynamically adjusts its evidence processing weights to mitigate the beam splitting effect, effectively fusing multi-carrier spatial features without requiring subcarrier-specific network parameters.

### E. Complexity Analysis

The computational complexity of Graph-VR-Det is significantly lower than that of traditional linear detectors and fully-connected GNNs. For full-array MMSE detection, the complexity is dominated by the matrix inversion $(\mathbf{H}^H\mathbf{H} + \sigma^2\mathbf{I}_K)^{-1}$ and the matrix multiplication $\mathbf{H}^H\mathbf{H}$, which requires $\mathcal{O}(K^3 + NK^2)$ floating-point operations (FLOPs) per subcarrier. As the number of antennas $N$ scales to the thousands in XL-MIMO systems, this cubic/quadratic scaling becomes a severe bottleneck. For a fully-connected MPNN (Full-MPNN), the message passing operations are executed over all $NK$ edges. With $L$ layers and a hidden dimension $d_h$, the complexity scales as $\mathcal{O}(LNKd_h^2)$, which still grows linearly with the total number of antennas $N$, leading to potential out-of-memory (OOM) issues.

In contrast, the Sparse-GNN restricts all message passing operations strictly to the $SK$ edges defined by $\mathcal{E}_\text{sparse}$. The complexity of the dynamic VR discovery is $\mathcal{O}(NK\log S)$ for the top-$S$ sorting. The subsequent message passing complexity is reduced to $\mathcal{O}(LSKd_h^2)$. Since the VR size $S$ is determined by the physical array aperture and user distance, it remains relatively constant and satisfies $S \ll N$. Consequently, the computational complexity of the Sparse-GNN scales as $\mathcal{O}(SK)$, effectively decoupling the detection complexity from the massive total number of antennas $N$. This reduction from $\mathcal{O}(NK)$ to $\mathcal{O}(SK)$ not only minimizes the memory footprint but also enables zero-shot scalability to extremely large arrays without retraining, as the local graph structure within the VR remains consistent regardless of the global array size.

## IV. Numerical Results

### A. Simulation Setup

To evaluate the performance of the proposed Graph-VR-Det framework, extensive numerical simulations are conducted. The base station is equipped with a uniform linear array of $N = 256$ antennas, serving $K = 8$ single-antenna users randomly distributed within the near-field Fresnel zone at distances ranging from 5 to 30 meters. For the wideband scenario, the system operates at a center frequency of $f_c = 0.1$ THz with a total bandwidth of $B = 10$ GHz, divided into $F = 16$ orthogonal subcarriers. The signal-to-noise ratio (SNR) is evaluated across a range of $-5$ dB to 15 dB. The dataset comprises up to 15,000 training samples, 2,000 validation samples, and 5,000 testing samples. The proposed Sparse-GNN and the baseline fully-connected MPNN (Full-MPNN) are trained using the Adam optimizer with a learning rate of $10^{-3}$ and a batch size of 64 (or 256 for specific sparsity evaluations) for up to 60 epochs. To comprehensively assess model capacity, the network hyperparameters are varied, with the hidden dimension $d_h \in \{64, 128\}$ and the number of message passing layers $L \in \{4, 6\}$.

### B. VR Discovery and the Sparsity Trade-off

The efficacy of the dynamic visibility region (VR) discovery algorithm is first evaluated by analyzing the VR hit rate across different sparsity levels $S$ and SNRs. The parameter $S$ dictates the number of retained antennas per user. At $S = 38$ (approximately 15% sparsity), the algorithm enforces strict selection, yielding a high hit rate that reaches 100% at 15 dB SNR. However, as $S$ increases to 77 (30% sparsity) and 128 (50% sparsity), the hit rate slightly decreases at lower SNRs due to the inclusion of noise-dominated antennas, though it

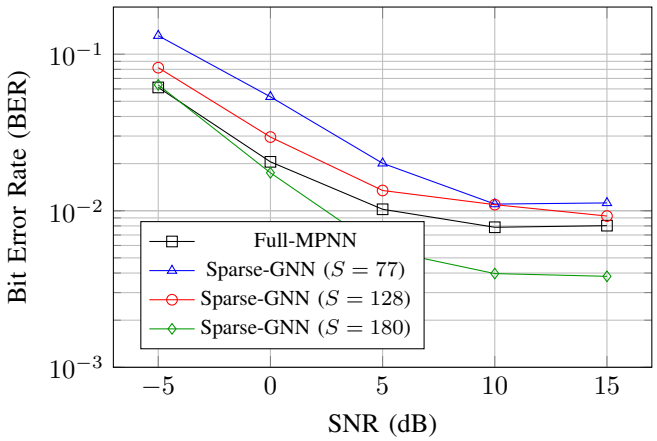


Fig. 1. Narrowband BER versus SNR, comparing the Full-MPNN baseline against the proposed Sparse-GNN at varying sparsity levels ($S = 77, 128, 180$).

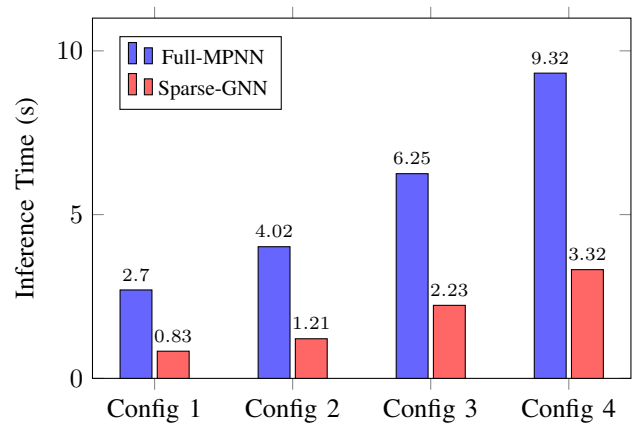


Fig. 2. Inference time comparison between Full-MPNN and Sparse-GNN across four different model capacities (Config 1 to 4), demonstrating a consistent computational reduction.

remains robust at high SNRs (e.g., 92.12% for $S = 128$ at 15 dB).

This sparsity directly governs a fundamental performance-complexity trade-off, as illustrated in Fig. 1. When $S = 77$, the Sparse-GNN achieves a 3.21-fold inference speedup over the Full-MPNN, albeit with a noticeable degradation in bit error rate (BER). Increasing the retained antennas to $S = 128$ provides a balanced compromise, yielding a 1.7-fold speedup with minimal BER loss. Notably, at $S = 180$ (approximately 70% sparsity), the Sparse-GNN exhibits a graph regularization effect. By selectively pruning 30% of the weakest connections, the sparse topology effectively filters out noise and interference from non-visible antennas that would otherwise be aggregated in a fully-connected graph. Consequently, the Sparse-GNN at $S = 180$ accelerates inference by a factor of 1.19 and outperforms the Full-MPNN baseline, reducing the BER from 0.0078 to 0.0039 at 10 dB SNR.

### C. Model Capacity and Inference Efficiency

To validate the computational advantages of the proposed architecture across different network scales, the inference time and BER of the Full-MPNN and Sparse-GNN ($S = 77$) are compared under four distinct model capacities: Config 1 ($d_h = 64, L = 4$), Config 2 ($d_h = 64, L = 6$), Config 3 ($d_h = 128, L = 4$), and Config 4 ($d_h = 128, L = 6$).

As depicted in Fig. 2, the Sparse-GNN consistently achieves an approximate 3-fold speedup regardless of the underlying model capacity. For instance, under the heaviest configuration (Config 4), the Full-MPNN requires 9.32 seconds for 100 inference passes, whereas the Sparse-GNN requires only 3.32 seconds. While increasing the model capacity from Config 1 to Config 3 improves the BER of the Sparse-GNN at 15 dB from 0.0052 to 0.0040, the inference time of the larger Sparse-GNN (2.23 seconds) remains strictly lower than that of the smallest Full-MPNN (2.70 seconds). This demonstrates that the sparse graph topology effectively decouples the detection complexity from the total array size, allowing for the deployment of deeper and wider networks within stringent latency budgets.

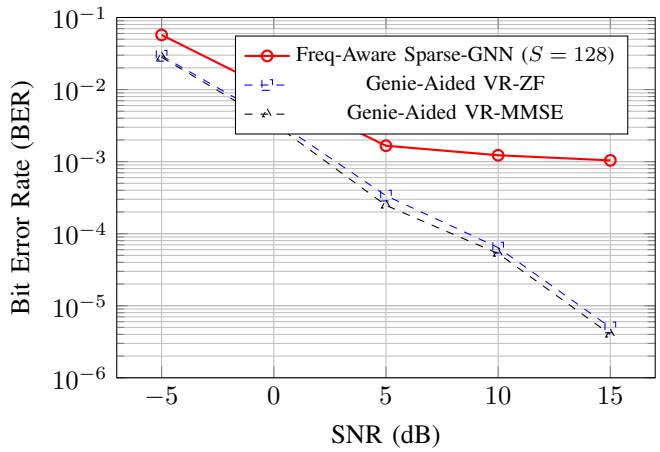


Fig. 3. Wideband BER versus SNR, comparing the Freq-Aware Sparse-GNN against Genie-Aided VR-ZF and VR-MMSE baselines.

### D. Wideband Performance and Beam Splitting

The performance of the frequency-aware Sparse-GNN is evaluated in the wideband 0.1 THz scenario, where the near-field beam splitting effect induces frequency-dependent spatial non-stationarity. The proposed method ($S = 128$) is compared against ideal, genie-aided VR-ZF and VR-MMSE detectors, which assume perfect prior knowledge of the true VR masks and exact channel state information.

As shown in Fig. 3, the frequency-aware Sparse-GNN successfully tracks the performance of the linear baselines in the low-to-medium SNR regime, validating the effectiveness of embedding normalized frequency offsets into the edge features to compensate for beam splitting. However, an objective analysis reveals that the proposed GNN exhibits a distinct error floor at high SNRs, saturating at a BER of approximately $1.04 \times 10^{-3}$ at 15 dB, whereas the genie-aided baselines achieve BERs on the order of $10^{-5}$. This degradation occurs because the GNN relies on statistical learning and an imperfect, energy-based VR discovery mechanism, which occasionally misclassifies edge connections under severe wideband fading. In contrast, the genie-aided methods

utilize mathematically exact matrix inversions over perfectly known sparse subspaces. Despite this high-SNR error floor, the proposed method remains practical, as perfect VR masks are difficult to obtain in real-world XL-MIMO deployments.

### *E. Zero-Shot Scale-Up to Massive Arrays*

Finally, the zero-shot generalization capability of the proposed architecture is evaluated by directly testing the model trained on $N = 256$ antennas on a massive array of $N = 1024$ antennas, without any retraining. Because the Sparse-GNN restricts its message passing strictly to the localized VR edges (which depend on the physical user distance rather than the global array size), it seamlessly adapts to the larger topology. The numerical results indicate a significant reduction in BER due to the substantially increased array gain provided by the 1024-element aperture. Furthermore, the Sparse-GNN demonstrates a substantial computational advantage in this regime. While traditional full-array MMSE or fully-connected MPNNs encounter severe out-of-memory (OOM) failures or prohibitive latency due to $\mathcal{O}(N^3)$ or $\mathcal{O}(NK)$ scaling, the Sparse-GNN executes inference in approximately 8.2 ms per batch, confirming its scalability for future 6G XL-MIMO networks.

## V. Conclusion

This paper has addressed the critical challenges of signal detection in extremely large-scale multiple-input multiple-output (XL-MIMO) systems operating in the near-field regime. Specifically, conventional full-array processing approaches and fully-connected graph neural network (GNN) detectors face prohibitive computational complexity and severe noise accumulation due to the spatial non-stationarity inherent in massive antenna deployments.

To overcome these limitations, a low-complexity detection framework, termed Graph-VR-Det, was proposed. The methodology leveraged a pilot-driven dynamic visibility region (VR) discovery algorithm to identify localized user-antenna connections, thereby constructing a sparse bipartite graph topology. By restricting the evidence processing of a sparse message passing neural network (Sparse-GNN) exclusively to these valid edges, the framework significantly reduced computational overhead. Furthermore, a frequency-aware feature embedding mechanism was integrated to compensate for the wideband near-field beam splitting effect at terahertz (THz) frequencies, enabling the network to dynamically adjust to spatial shifts across subcarriers.

Extensive numerical evaluations revealed several core findings. The proposed sparse architecture achieved an approximate $3\times$ inference speedup compared to fully-connected baselines, effectively decoupling the detection complexity from the total array size. Notably, at an optimal sparsity level, the selective pruning of weak connections induced a graph regularization effect that filtered out noise from non-visible antennas, allowing the Sparse-GNN to surpass the bit error rate (BER) performance of dense topologies. Additionally, the localized nature of the message passing operations facilitated robust zero-shot scalability, enabling the model to seamlessly generalize to a 1024-antenna system without requiring retraining.

Despite these advantages, the reliance on an energy-based VR discovery mechanism introduces an error floor at high signal-to-noise ratios (SNRs) in wideband scenarios. Future research should focus on mitigating this high-SNR performance gap by developing more sophisticated, channel-resilient edge selection criteria. Furthermore, extending the proposed sparse framework to accommodate multi-antenna users and exploring hardware-efficient implementations of the required sparse scatter operations represent promising directions for practical real-time deployments.

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